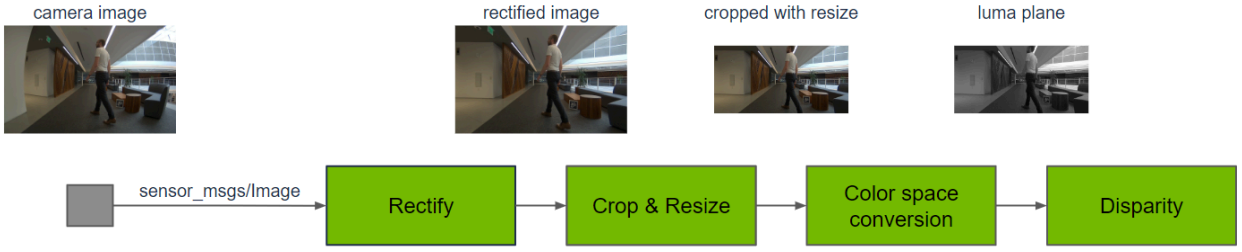
Isaac ROS Image distortion processing

Isaac ROS Image distortion processing
- Overview
- Quick experience

Isaac ROS Image Pipeline official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_image_pipeline/index.html

Overview

Isaac ROS Image Pipeline provides image preprocessing functions, including cropping, scaling, lens distortion correction, flipping and colour space conversion. For stereo cameras it can also generate disparity and point clouds from the left and right images.



Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Each terminal has to use the same ROS/FastDDS environment. The processing chain takes `/image` and `/camera_info` as the input of `RectifyNode`, outputs `/image_rect` and `/camera_info_rect`, and then `ResizeNode` outputs `/resize/image`.

Note: all the terminals below are executed inside the container:

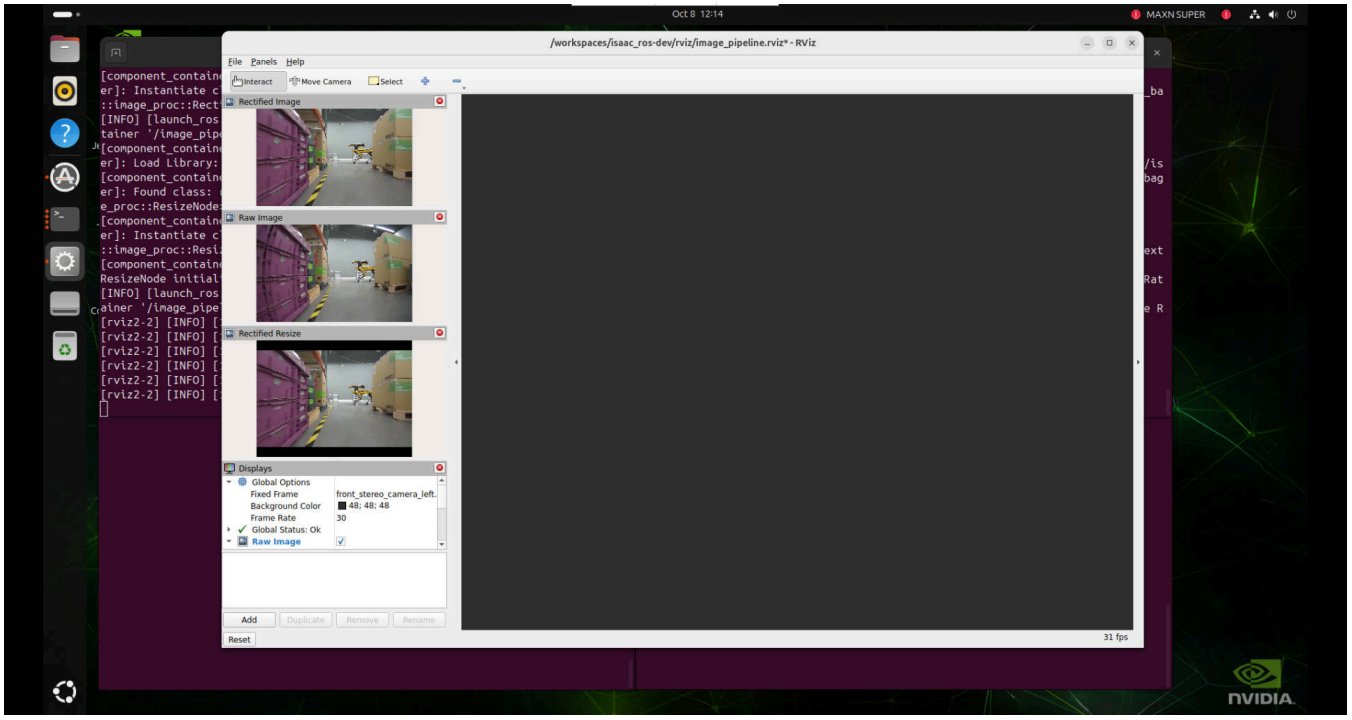
Terminal A: start the processing chain and RViz:

```
ros2 launch /workspaces/isaac_ros-dev/launch/image_pipeline.launch.py
```

The RViz configuration is `/workspaces/isaac_ros-dev/rviz/image_pipeline.rviz`, which displays `/image`, `/image_rect` and `/resize/image`.

Terminal B: play the input bag in a loop:

```
ros2 bag play --loop --rate 0.5 \
  /workspaces/isaac_ros-dev/isaac_ros_assets/isaac_ros_visual_slam/quickstart_bag \
  --remap \
  /front_stereo_camera/left/image_raw:=/image \
  /front_stereo_camera/left/camera_info:=/camera_info
```



Terminal C: check the input, the undistorted output and the scaled output:

```
ros2 topic echo --once /image --field width
ros2 topic echo --once /image_rect --field width
ros2 topic echo --once /resize/image --field width
ros2 topic echo --once /camera_info_rect --field d
```